

Data Flow Diagram

Date	25 March 2026
Team ID	xxxxxx
Project Name	OptiCrop – AI-Based Crop Recommendation System
Maximum Marks	2 Marks

Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how data flows through the OptiCrop – AI-Based Crop Recommendation System. The farmer enters soil nutrient values and environmental parameters through the web application. Finally, the predicted crop recommendation is displayed to the farmer, helping them make informed agricultural decisions.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	Farmer/User who provides soil and environmental details and receives crop recommendations.
Rectangle with numbered header	Process	Receives user input, preprocesses data, performs Machine Learning prediction, and generates the recommended crop.
Rectangle	Data Store	Crop Recommendation Dataset, Trained Machine Learning Model (model.pkl), and Scaler (scaler.pkl).
Arrow	Data Flow	Movement of soil data, environmental parameters, prediction request, and recommended crop between entities, processes, and data stores.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

